

HOLLISTER INCORPORATED

Topic Naming Convention Standard

ADDENDUM A

Unified Messaging Taxonomy from PLC to ERP

The Traceability Imperative

A message must be traceable from a specific sensor on the shop floor all the way to a financial record in the cloud without losing context.

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1. Introduction

1.1 Purpose

This addendum establishes the mandatory naming conventions for all messaging topics across Hollister's Digital Factory architecture. It defines how data flows from PLCs and sensors (MQTT) through the integration bridge to enterprise systems (Kafka), ensuring complete traceability and semantic consistency.

The conventions balance two competing requirements:

- MQTT's hierarchical flexibility for edge-level control and monitoring with wildcard subscriptions
- Kafka's flat, high-throughput structure optimized for enterprise processing and analytics

1.2 Scope

This standard applies to:

- All MQTT topics published by PLCs, sensors, edge gateways, and MES systems
- All Kafka topics consumed by enterprise applications (ERP, CRM, PLM, SCM, Analytics)
- All bridge configurations that map MQTT topics to Kafka topics
- All new integrations and system deployments across global manufacturing facilities

1.3 Design Principles

Principle	Description
Physical-to-Logical	MQTT topics mirror physical plant hierarchy; Kafka topics organize by business domain
Traceability	Every message can be traced from sensor to financial record without losing context
Consistency	Single casing convention (snake_case), no mixed formats across the enterprise
Versioning	Kafka topics include version suffix to enable schema evolution without breaking consumers
State Over Action	Topics describe state or identity, not actions; commands are payload content
Context Preservation	Bridge enriches messages with metadata; original MQTT hierarchy becomes Kafka message key

2. MQTT Topic Naming Convention (Edge Layer)

2.1 The Physical-to-Logical Hierarchy

MQTT topics follow a hierarchical structure that mirrors the physical plant layout, aligned with ISA-95 equipment hierarchy. This enables powerful wildcard subscriptions for localized monitoring and control.

2.1.1 Standard Template

```
[enterprise]/[site]/[area]/[line]/[cell]/[device]/[metric_category]/[metric]
```

Components:

└─ enterprise		Organization identifier (hollister)
└─ site		Physical facility (bal, sdr, lib, kau, fre)
└─ area		Production area within site (mh1, pkg, cln, asm)
└─ line		Production line (line_01, line_02)
└─ cell		Work cell or station (cell_a, station_01)
└─ device		Equipment identifier (mold_01, press_02, robot_03)
└─ metric_category		Data category (telemetry, production, quality, alarms, state)
└─ metric		Specific measurement (temperature, pressure, cycle_count)

2.1.2 Hierarchy Level Definitions

Level	ISA-95 Equiv.	Format	Examples
Enterprise	Enterprise	Lowercase, single word	hollister
Site	Site	3-letter code, lowercase	bal (Ballina), sdr (Stuarts Draft), lib (Libertyville)
Area	Area	Abbreviated, snake_case	mh1 (Molding Hall 1), pkg (Packaging), cln (Cleanroom)
Line	Work Center	line_##, snake_case	line_01, line_02, cath_line_01
Cell	Work Unit	cell_X or station_##	cell_a, cell_b, station_01
Device	Equipment	type_##, snake_case	mold_01, press_02, robot_03, sensor_temp_01
Metric Cat.	-	Fixed vocabulary	telemetry, production, quality, alarms, state, maintenance
Metric	-	snake_case	temperature, pressure, cycle_count, part_present

2.2 Complete MQTT Topic Examples

MQTT Topic	Description
hollister/bal/mh1/line_01/cell_a/mold_01/telemetry/temperature	Mold temperature reading
hollister/bal/mh1/line_01/cell_a/mold_01/telemetry/pressure	Hydraulic pressure reading
hollister/bal/mh1/line_01/cell_a/mold_01/production/cycle_complete	Cycle completion event
hollister/bal/mh1/line_01/cell_a/mold_01/production/part_count	Parts produced counter
hollister/bal/mh1/line_01/cell_a/mold_01/quality/inspection_result	Inline inspection pass/fail
hollister/bal/mh1/line_01/cell_a/mold_01/alarms/active	Currently active alarms
hollister/bal/mh1/line_01/cell_a/mold_01/state/packml	Equipment state (PackML)
hollister/sdr/pkg/line_02/station_01/labeler_01/production/ label_applied	Label application event

2.3 Wildcard Subscription Patterns

MQTT wildcards enable flexible subscription patterns for different monitoring needs:

Subscription Pattern	Use Case
hollister/bal/#	All data from Ballina facility (plant manager view)
hollister/bal/mh1/line_01/#	All data from Line 01 (line supervisor view)
hollister/+/+/+/+/alarms/active	All active alarms globally (alarm management system)
hollister/bal/+/+/+/state/packml	All equipment states at Ballina (OEE dashboard)
hollister/+/+/+/mold_+/telemetry/#	All telemetry from all molds globally (predictive maintenance)
hollister/bal/mh1/line_01/cell_a/mold_01/#	All data from specific machine (Digital Twin)

3. Kafka Topic Naming Convention (Enterprise Layer)

3.1 The Functional Domain Structure

Kafka topics organize data by business domain and function rather than physical hierarchy. This structure optimizes for enterprise processing patterns where consumers need all data of a particular type regardless of source.

3.1.1 Standard Template

```
[environment].[domain].[subdomain].[data_type].[version]
```

Components:

|

— environment	Deployment environment (prod, staging, dev)
— domain	Business domain (factory, supplychain, quality, finance)
— subdomain	Functional area within domain (telemetry, orders, inspections)
— data_type	Type of data or event (raw, aggregated, enriched, commands)
— version	Schema version (v1, v2, v3)

3.1.2 Domain Level Definitions

Level	Format	Valid Values
Environment	Lowercase, single word	prod, staging, dev, test
Domain	Lowercase, business area	factory, supplychain, quality, finance, customer, maintenance
Subdomain	Lowercase, functional	telemetry, production, orders, inspections, inventory, alarms, equipment
Data Type	Lowercase, data nature	raw (unprocessed), enriched (with context), aggregated (summarized), commands (southbound)
Version	v# format	v1, v2, v3 (increment when schema changes breaking compatibility)

3.2 Complete Kafka Topic Examples

Kafka Topic	Partitions	Consumers
prod.factory.telemetry.raw.v1	By equipment_id (50)	Historian, AI/ML
prod.factory.production.enriched.v1	By work_order (20)	ERP, MES Analytics
prod.factory.equipment.state.v1	By site (5)	OEE Dashboard, ERP
prod.quality.inspections.enriched.v1	By product (20)	QMS, Regulatory

prod.supplychain.inventory.raw.v1	By warehouse (10)	ERP, SCM
prod.maintenance.alarms.enriched.v1	By severity (3)	CMMS, Analytics
prod.customer.orders.enriched.v1	By customer (50)	CRM, Portal
prod.factory.equipment.commands.v1	By equipment_id (50)	Bridge (southbound)

3.3 Why Flat Topics in Kafka?

Unlike MQTT, Kafka does not support hierarchical wildcards effectively. Flattening topics provides:

- Optimized Throughput: Kafka performs best with fewer, high-volume topics rather than millions of sparse topics
- Consumer Groups: All data of a type can be consumed by a single consumer group with parallel partitions
- Compaction: Log compaction works per-topic, enabling efficient state stores
- Retention Policies: Different data types can have different retention (telemetry: 7 days, quality: 2 years)
- Schema Management: One schema per topic in Schema Registry simplifies governance

4. The Mapping Strategy: Bridging MQTT to Kafka

4.1 Transformation Rules

The Integration Bridge performs critical transformations when moving data from MQTT to Kafka. The original MQTT hierarchy is preserved as metadata, ensuring traceability.

Element	MQTT (Source)	Kafka (Destination)
Topic Name	hollister/bal/mh1/line_01/cell_a/mold_01/telemetry/temperature	prod.factory.telemetry.raw.v1
Message Key	(Usually null or client ID)	bal:mh1:line_01:cell_a:mold_01
Payload	{"value": 185.2, "unit": "degC"}	Avro record with full metadata (see below)
Timestamp	Broker timestamp or payload	event_time (original) + kafka_time (ingestion)

4.2 Message Key Construction

The Kafka message key is constructed from the MQTT topic hierarchy using colon separators. This ensures:

- **Partition Affinity:** All messages from the same equipment land in the same partition (ordering guarantee)
- **Traceability:** Key decodes back to physical source without parsing payload
- **Compaction:** Log compaction can keep latest state per equipment

Key Construction Formula:

MQTT: hollister/{site}/{area}/{line}/{cell}/{device}/{category}/{metric}

Key: {site}:{area}:{line}:{cell}:{device}

Examples:

MQTT Topic: hollister/bal/mh1/line_01/cell_a/mold_01/telemetry/temperature

Kafka Key: bal:mh1:line_01:cell_a:mold_01

MQTT Topic: hollister/sdr/pkg/line_02/station_01/labeler_01/production/
label_applied

Kafka Key: sdr:pkg:line_02:station_01:labeler_01

4.3 Payload Enrichment

The bridge enriches the minimal MQTT payload with metadata required for enterprise processing:

MQTT Payload (Input):

```
{  
  "value": 185.2,  
  "unit": "degC"  
}
```

Kafka Payload (Output - Avro):

```
{  
  "event_id": "550e8400-e29b-41d4-a716-446655440000",  
  "event_time": 1708122000000,  
  "kafka_time": 1708122000150,  
  
  "source": {  
    "enterprise": "hollister",  
    "site": "bal",  
    "area": "mh1",  
    "line": "line_01",  
    "cell": "cell_a",  
    "device": "mold_01",  
    "metric_category": "telemetry",  
    "metric": "temperature",  
    "mqtt_topic": "hollister/bal/mh1/line_01/cell_a/mold_01/telemetry/temperature"  
  },  
  
  "data": {  
    "value": 185.2,  
    "unit": "degC",  
    "quality": "good"  
  },  
  
  "context": {
```

4.4 Topic Mapping Matrix

This matrix defines how MQTT topic patterns map to Kafka topics:

MQTT Topic Pattern	Kafka Topic
hollister/+/+/+/+/telemetry/#	prod.factory.telemetry.raw.v1
hollister/+/+/+/+/production/#	prod.factory.production.enriched.v1
hollister/+/+/+/+/state/#	prod.factory.equipment.state.v1
hollister/+/+/+/+/quality/#	prod.quality.inspections.enriched.v1
hollister/+/+/+/+/alarms/#	prod.maintenance.alarms.enriched.v1
hollister/+/+/+/+/maintenance/#	prod.maintenance.events.enriched.v1

5. Naming Rules and Best Practices

5.1 Mandatory Naming Rules

RULE 1: Use snake_case Consistently

All topic names, at all levels, must use snake_case (lowercase with underscores). MQTT is case-sensitive; mixing cases causes subscription failures.

✓ CORRECT	✗ INCORRECT
hollister/bal/mh1/line_01/mold_01	Hollister/BAL/MH1/Line01/Mold01
prod.factory.telemetry.raw.v1	prod.Factory.Telemetry.Raw.V1
cycle_complete	cycleComplete, CycleComplete, cycle-complete

RULE 2: Version All Kafka Topics

Every Kafka topic must include a version suffix (e.g., .v1). This enables schema evolution without breaking existing consumers. Never change a schema in place; create a new version.

Schema Evolution Example:

Version 1 (Original):

```
Topic: prod.factory.production.enriched.v1

Schema: { event_id, timestamp, equipment_id, quantity }
```

Version 2 (Added field):

```
Topic: prod.factory.production.enriched.v2

Schema: { event_id, timestamp, equipment_id, quantity, cycle_time }  <- New field
```

Migration Strategy:

- 1. Deploy new consumers that handle v2
- 2. Configure bridge to dual-write to v1 AND v2
- 3. Migrate consumers from v1 to v2
- 4. Deprecate v1 topic after all consumers migrated

RULE 3: Topics Describe State, Not Actions

Topic names should describe WHAT the data is, not what to DO with it. Commands are payload content, not topic names. This prevents tight coupling between publishers and subscribers.

✓ CORRECT (State/Identity)	✗ INCORRECT (Action)
hollister/bal/.../mold_01/state/packml	hollister/bal/.../mold_01/commands/stop
Payload: {"state": "stopping", "reason": "operator"}	(action in topic name)
prod.factory.equipment.commands.v1	prod.factory.equipment.stop_machine.v1
Payload: {"command": "stop", "target": "mold_01"}	(action in topic name)

5.2 Additional Best Practices

Practice	Rationale
----------	-----------

Keep MQTT topics granular	One metric per topic enables precise subscriptions; combine in Kafka, not MQTT
Keep Kafka topics coarse	Fewer topics with more partitions scales better; use message key for routing
Use fixed vocabulary for categories	Standardize on: telemetry, production, quality, alarms, state, maintenance, commands
Include full path in Kafka payload	Store original MQTT topic in payload for debugging and lineage tracking
Document all topics in registry	Maintain a central topic registry with owners, schemas, retention policies, consumers
Avoid special characters	Only use alphanumeric, underscore, period (Kafka), slash (MQTT); no spaces, hyphens in topics

6. End-to-End Example: Part Produced Event

6.1 Scenario

A molding machine at Ballina completes a production cycle, producing one good part. This event must flow from the PLC to the ERP for inventory update and to the CRM for customer order status.

6.2 Step-by-Step Flow

STEP 1: PLC Publishes to MQTT

Topic: hollister/bal/mh1/line_01/cell_a/mold_01/production/cycle_complete

QoS: 1 (at least once)

Payload: {

"count": 1,

"cycle_time_sec": 12.3,

"result": "good"

}

STEP 2: MES Receives via MQTT (Real-time)

Subscription: hollister/bal/mh1/line_01/#

Action:

- Updates work order WO-2026-001234 progress
- Records cycle time in eDHR
- Adds context: work_order, material_lot, operator

STEP 3: Bridge Transforms and Publishes to Kafka

Topic: prod.factory.production.enriched.v1

Key: bal:mh1:line_01:cell_a:mold_01

Payload (Avro):

{

"event_id": "550e8400-e29b-41d4-a716-446655440000",

"event_time": 1708122000000,

"event_type": "CYCLE_COMPLETE",

"source": {

6.3 Traceability Verification

At any point in this flow, we can trace the message back to its source:

Query	Answer (from Kafka message)
Which machine?	mold_01 (from key: bal:mh1:line_01:cell_a:mold_01)
Which line?	line_01 in Molding Hall 1, Ballina (from source.line, source.area, source.site)
What product?	CATH-001 (from context.product_code)
Which work order?	WO-2026-001234 (from context.work_order)
Which material lot?	LOT-2026-0456 (from context.material_lot)
Original MQTT topic?	hollister/bal/mh1/line_01/cell_a/mold_01/production/cycle_complete (from source.mqtt_topic)
When did it happen?	2024-02-17T00:00:00Z (from event_time)

7. Topic Registry Template

7.1 MQTT Topic Registry

All MQTT topics must be registered in the central topic registry. Template:

Field	Example Entry
Topic Pattern	hollister/{site}/{area}/{line}/{cell}/{device}/telemetry/temperature
Description	Equipment barrel/mold temperature readings from injection molding machines
Publisher	Edge Gateway (Kepware) via OPC-UA to MQTT
QoS	0 (at most once)
Frequency	Every 1 second per device
Payload Schema	{"value": float, "unit": string, "quality": string}
Subscribers	MES (real-time), Digital Twin, Bridge (to Kafka)
Owner	Manufacturing Systems Team
Kafka Mapping	prod.factory.telemetry.raw.v1

7.2 Kafka Topic Registry

All Kafka topics must be registered with schema and retention policies. Template:

Field	Example Entry
Topic Name	prod.factory.production.enriched.v1
Description	Enriched production events from all manufacturing equipment globally
Partitions	50 (partitioned by equipment_id)
Replication Factor	3
Retention	90 days (log retention), 7 days (compacted)
Schema ID	factory.production.enriched-v1 (in Schema Registry)
Key Format	String: {site}:{area}:{line}:{cell}:{device}
Producers	Integration Bridge (Kafka Connect)
Consumer Groups	erp-production-consumer, crm-order-status-consumer, plm-analytics-consumer, historian-archiver
Owner	Integration Platform Team
MQTT Sources	hollister/+/+/+/+/+/production/#

Appendix A: Quick Reference Card

MQTT Topic Template

[enterprise]/[site]/[area]/[line]/[cell]/[device]/[metric_category]/[metric]

Example: hollister/bal/mh1/line_01/cell_a/mold_01/telemetry/temperature

Kafka Topic Template

[environment].[domain].[subdomain].[data_type].[version]

Example: prod.factory.telemetry.raw.v1

Kafka Message Key Template

[site]:[area]:[line]:[cell]:[device]

Example: bal:mh1:line_01:cell_a:mold_01

Site Codes

Code	Site	Location
bal	Ballina Manufacturing	Ballina, County Mayo, Ireland
sdr	Stuarts Draft Manufacturing	Stuarts Draft, Virginia, USA
lib	Libertyville HQ	Libertyville, Illinois, USA
kau	Kaunas Manufacturing	Kaunas, Lithuania
fre	Frederiksberg Manufacturing	Frederiksberg, Denmark

Metric Categories (Fixed Vocabulary)

Category	Content
telemetry	Sensor readings: temperature, pressure, vibration, speed, flow, humidity
production	Production events: cycle_complete, part_count, reject_count, batch_start, batch_end
quality	Quality data: inspection_result, spc_measurement, ncr_created, hold_status
alarms	Alarm events: active, acknowledged, cleared, suppressed
state	Equipment state: packml (PackML states), mode, availability

maintenance	Maintenance events: request, scheduled, in_progress, complete
commands	Southbound commands: setpoint, recipe, start, stop (payload contains action)

End of Addendum A: Topic Naming Convention Standard